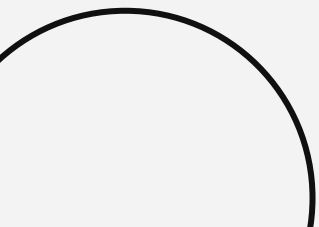




Research Title:

Impact of heatwaves on human health;

A Case Study in Pabna



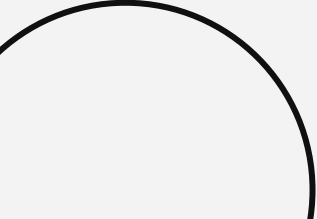


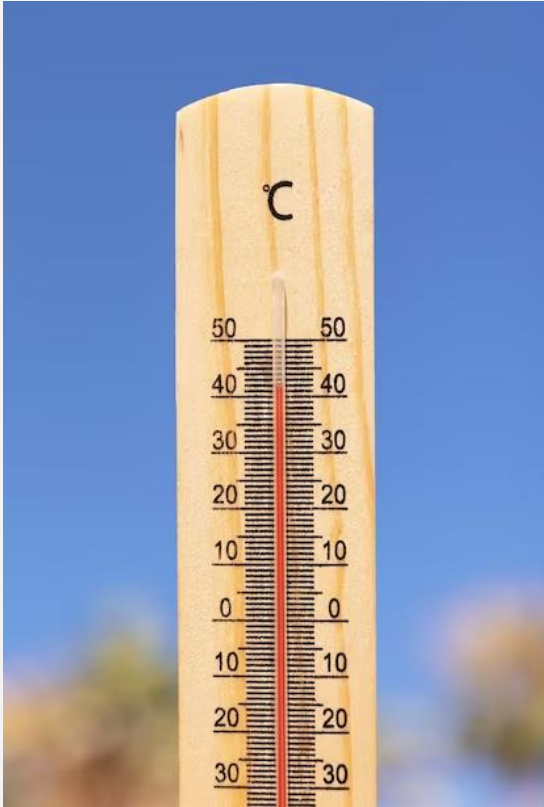
Outline:

- ❖ Abstract
- ❖ Introduction
- ❖ Literature Review
- ❖ Research Gap
- ❖ Objectives
- ❖ Methodology
- ❖ Expected Outcome



A Case Study in Pabna





What is Heatwaves ?

A prolonged period of abnormal hot weather known as heatwaves



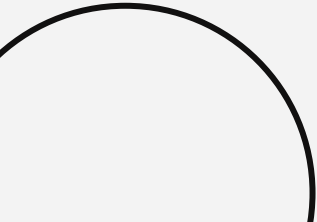
A Case Study in Pabna





Abstract:

This research explores the impact of heatwaves on human health in Pabna City, Bangladesh. Through a cross-sectional study, data was collected from 350 individuals using structured questionnaires and in-depth interviews. The study reveals the physical and mental health consequences of heatwaves, coping strategies, and socio-economic factors influencing vulnerability. Findings contribute valuable insights to the existing knowledge on heatwave-related health risks and offer recommendations for targeted interventions to protect public health during extreme heat events.



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Introduction

Heatwaves

A prolonged period of abnormal hot weather, known as heatwaves, has become an increasingly prevalent and concerning global climate phenomenon. Heatwaves have severe implications for human health and well-being, posing significant challenges to vulnerable populations, particularly in urban centers. As the earth's temperatures rise due to **climate change**, the frequency, intensity, and duration of heatwaves are expected to increase, making them a critical public health concern.



Why pabna city-

One such vulnerable urban center is Pabna City, located in the northwest region of Bangladesh. This densely populated city experiences intense heatwaves during certain periods of the year, impacting the health and daily lives of its residents. As the effects of climate change continue to unfold, **understanding the specific risks and impacts of heatwaves on human health** in Pabna City is of absolute importance.



Introduction:



The impact of heatwaves on human health is a pressing research area that demands **urgent attention** to develop effective strategies to safeguard public health during extreme heat events. Previous studies have shown that heatwaves can lead to a range of health issues, from mild heat-related illnesses like heat rashes to severe conditions such as **heat exhaustion and heatstroke**. Additionally, heatwaves have been associated with exacerbating existing health problems, including **respiratory and cardiovascular diseases, kidney disorders, and mental health challenges**.

While the health impacts of heatwaves are evident, the vulnerability and coping strategies of individuals in Pabna City require **further exploration**. The unique geographical context and local socio-economic factors may influence the health effects experienced by the city's diverse population. Understanding these dynamics will enable the development of targeted interventions and policies to protect vulnerable groups effectively.

This research proposal aims to investigate the "Impact of Heatwaves on Human Health: A Study in Pabna City, Bangladesh." By delving into the health effects of heatwaves in this specific urban context, the study endeavors to contribute to the existing body of knowledge on climate change-related health risks, particularly concerning extreme temperatures.

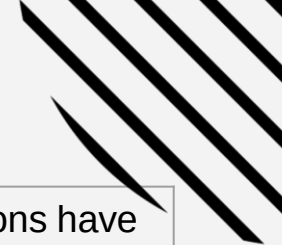
Literature Review :



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Literature Review:

The literature review provided highlights some important findings related to the impact of heatwaves on human health



Anderson and Bell (2011)	▪ Several studies across different geographical locations have shown a clear association between heatwaves and increased mortality and morbidity.
HEAT WAVES AND HUMAN HEALTH (February 2019)	▪ The impacts of heatwaves on human health are influenced by the vulnerability and exposure of the affected population. ▪ Vulnerable populations may experience more significant health risks during heatwaves
WHO(World Health Organization) 1 June 2018	▪ Rising global ambient temperatures affect all populations. However, some populations are more exposed to, or more physiologically or socio-economically vulnerable to physiological stress, exacerbated illness, and an increased risk of death from exposure to excess heat. These include the elderly, infants and children, pregnant women, outdoor and manual workers, athletes, and the poor. Gender can play an important role in determining heat exposure





Literature Review:

The literature review provided highlights some important findings related to the impact of heatwaves on human health

Klinenberg (2002)	The relationship between temperature and health impacts is not uniform, and vulnerability varies across different sub-groups
Uejio et al. (2011), and Yihan et al. (2013)	*The complex and interacting factors, including biological, environmental, medical, social, and geographical, that influence the health effects of heatwaves on different populations.
(The impact of heat waves on mortality in Northwest India, Environmental Research, Volume 176, September 2019, 108546).	The impact of such events can be disastrous, especially in regions where the population lacks adequate preparedness to cope with high temperatures



Research Gap:

The research gap in this study is the lack of comprehensive understanding and investigation

☐ Geographical Context and Local Vulnerability

Most of the existing research on the health impacts of heatwaves has been conducted in developed countries or larger urban areas. There is a lack of studies that focus on the unique characteristics of smaller cities or urban centers in developing countries like Pabna City, Bangladesh. The research gap lies in the need to understand how the geographical context and local vulnerability factors influence the health effects of heatwaves in this specific city.

☐ Inadequate understanding of how socio-economic factors contribute to health disparities during heatwaves in Pabna City.





Research Gap:

The research gap in this study is the lack of comprehensive understanding and investigation

- ❑ Lack of knowledge on the effectiveness and feasibility of heatwave adaptation strategies and community resilience in this specific context.
- ❑ Insufficient exploration of heatwave awareness and behavior change among residents of Pabna City.
- ❑ Scarcity of studies examining the long-term health impacts of recurrent heatwave exposure in the region.



Objectives:

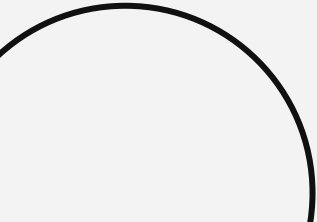
- To find out the impacts of heatwaves on human physical health and mental health.
- To determine the change of food habit during heatwaves.
- To find out the change of work performance and work productivity during heatwave.
- To assess the financial stability of outdoor Workers during heatwaves.





Methodology:

Research Design	This study will employ a cross-sectional research design to collect data at a specific point in time, capturing the impact of recent heatwaves on human health in Pabna City, Bangladesh. Both quantitative and qualitative data will be gathered to gain a comprehensive understanding of the subject.
Study Area	The study will be conducted in Pabna City, Bangladesh, due to its vulnerability to heatwaves and the relevance of the research topic to its urban population.



A Case Study in Pabna





Methodology:

Sample Selection

A sample size of 350 individuals between the ages of 15 and 50 years will be selected through simple random sampling to ensure representation across various age groups. The participants will include outdoor workers, businessmen, construction workers, rickshaw pullers, and others from different socio-economic backgrounds





Methodology

Data Collection:

Quantitative Data Collection:

A structured questionnaire will be used to collect quantitative data, covering demographics, physical health issues experienced during heatwaves, mental health impacts, changes in eating and sleeping patterns, water consumption, work performance, and financial challenges. Data on participants' monthly income and access to cooling facilities will also be collected.

Qualitative Data Collection:

In-depth interviews will be conducted with a subset of participants to gain deeper insights into their experiences and perceptions regarding heatwave impacts on health, coping strategies, and access to support or accommodations.



Methodology

Data Analysis:

Quantitative Data Analysis:

Statistical software will be used to analyze the quantitative data, employing descriptive statistics to summarize demographic characteristics and health impacts. Relationships between variables will be examined using correlation analysis. Any significant differences between subgroups will be assessed through inferential statistics (e.g., t-tests, ANOVA).

Qualitative Data Analysis:

Thematic analysis will be used to identify patterns and themes in the qualitative data obtained from interviews. The analysis will involve coding and categorizing participants' responses to explore common experiences and perspectives.
Ex-(chai-square test)



Methodology

Ethical Considerations:

Informed consent will be obtained from all participants before data collection, ensuring their voluntary participation.
Participants' privacy and confidentiality will be strictly maintained throughout the research process.

Limitations:

The cross-sectional design may limit the ability to **establish causal relationships between heatwaves and health outcomes**.
Self-reporting in the questionnaire and interviews may be subject to recall and social desirability biases.





Expected Outcome of The Research:

- Identification of Heatwaves Related Health Impacts.(Dehydration, Heatstroke)
- Understanding Coping Strategies.
- Socio-Economic Factors and Vulnerability.
- Impact on Work Performance and Productivity.
- Financial Challenges.



Outline:



Any Question?

A Case Study in Pabna



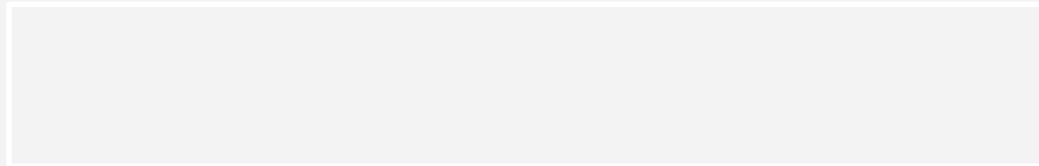
Thanks!

Do you have any questions?

Nazmulhossen@gmail.com

01835226777

<https://www.linkedin.com/in/nazmul-al-hossen-9b4b7a1a2/>





Our Team:

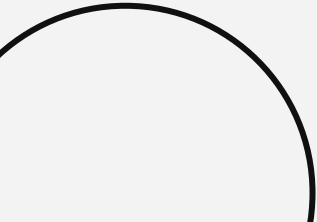
❖ Nazmul Hossen(181632)

Session: 2017-18

4th Year 1st Semester

Dept. of Statistics

Pabna University of Science and Technology



A Case Study in Pabna

